1. Clinical Question & Core Hypothesis

Clinical Question

MSH6 loss should increase neoantigen load and recruit TILs, which ought to slow progression, yet many MSH6-deficient bladder cancers still progress or recur. Why does tumor-intrinsic immunogenicity (MSH6 deficiency) fail to translate into durable tumor control in some patients? 2) Tier 1 paths used: I built on the MSH6-as-prognostic-factor path (links DNA repair → neoantigens → immune recognition) and the PNI path (host nutritional/immune competence), reasoning that the interaction between tumor immunogenicity and systemic immune fitness determines whether TILs are functionally protective. 3) Tier 2 support: AI-based biomarkers and EV/MDK signals suggest multi-modal assays can capture functional immune states and tumor aggressiveness; metabolic/urinary markers and MDK indicate host metabolic/inflammatory milieu and tumor-secreted factors can modulate immune efficacy, reinforcing the plausibility that systemic status (PNI) and tumor-derived suppressive signals alter TIL function. 4) Endpoint choice: I selected Progression-Free Interval (PFI) because the ontologist's synthesis focused on recurrence/progression risk and PFI is most sensitive to early failure driven by inadequate immune control; it aligns with clinical decisions about adjuvant therapy. 5) Deliberate exclusions: GREM1 (Tier 1) was excluded from the core hypothesis to keep the mechanistic question sharp (only 1 imaging feature + up to 2 anchors); GREM1 and other Tier2 nodes (MDK, EV mRNA, urinary metabolites, AI signatures) are listed as secondary_modifiers for follow-up analyses rather than conflating the primary causal claim?

Working Hypothesis

MSH6-deficient bladder tumors generate high neoantigen loads that recruit tumor-infiltrating lymphocytes (TILs). However, durable tumor control requires robust systemic immune/nutritional competence as captured by the Prognostic Nutritional Index (PNI). Specifically, among MSH6-deficient, TIL-high bladder cancers, individuals with low PNI exhibit impaired TIL cytotoxic function (reduced granzyme B and perforin, diminished IFN-γ signaling, and elevated exhaustion markers such as PD-1/LAG-3), leading to immune escape and shorter progression-free interval (PFI). In contrast, patients with preserved or high PNI maintain functional TIL activity and experience longer PFI. Tumor-derived modulators (e.g., MDK) and extracellular vesicle (EV) cargo, along with host urinary/metabolic signals, act as secondary modifiers that can further suppress or sustain TIL function, accounting for cases where high TIL density does not predict favorable outcomes. The core testable predictions are: (a) a significant interaction between TIL density (or CD8+ TIL metrics) and PNI on PFI in MSH6-deficient tumors; (b) TIL functional state correlates with PNI independent of TIL density; (c) MDK/EV/urinary metabolite signatures add predictive value beyond TIL density and PNI; and (d) interventions that improve nutritional/immunometabolic status or inhibit tumor-derived suppressive signals can shift at-risk patients toward longer PFI.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, PNI
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6 deficiency) fail to translate into durable tumor control in some patients? 2) Tier 1 paths used: I built on the MSH6-as-prognostic-factor path (links DNA repair → neoantigens → immune recognition) and the PNI path (host nutritional/immune competence), reasoning that the interaction between tumor immunogenicity and systemic immune fitness determines whether TILs are functionally protective.

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Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	GREM1	Supporting modifier from knowledge graph
Secondary	MDK	Supporting modifier from knowledge graph
Secondary	EV_mRNA_markers	Supporting modifier from knowledge graph
Secondary	AI-derived_biomarker_signatures	Supporting modifier from knowledge graph
Secondary	urinary_metabolites (creatinine, microalbumin, GLU)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- GREM1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MDK: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.